

Samarth Prabhu

ML Engineer | GenAI, Computer Vision & Applied ML

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PROFESSIONAL SUMMARY

ML Engineer with 5+ years of experience across applied **Machine Learning, Computer Vision, GenAI** and **data science**, currently building production AI workflows for a global e-commerce marketplace. Proposed and pitched **12+ ML/GenAI POCs** to architects, product teams and VP leadership, with **9+ approved**. Built 7+ production features across 3 AI services, including **product-data-validation** and **product-image processing**, that reduced product onboarding time by **40%**. Hands-on in Python, PyTorch, FastAPI, GCP and LangGraph.

WORK EXPERIENCE

Senior ML/AI Engineer

Dec 2024 – Present

Infosys, Pune | Client: Global E-commerce Marketplace

- Built and deployed an event-driven **product-data-validation service** augmenting manual review, combining **multimodal embeddings, classification, OCR verification** and **object detection** to flag incorrect listings for seller correction; achieved **~96%+ recall** against QA-reviewed (**60-70%**) production sample, improving listing quality.
- Engineered **image processing pipeline** using GCP (Cloud Pub/Sub-triggered, Cloud Storage, CPU-only inference) with BiRefNet segmentation, shadow preservation and image standardization to **generate catalog-ready images**, replacing an outsourced vendor serving as the **major contributor** to **40% reduction** in product onboarding time.
- Developed a **studio-vs-lifestyle** routing classifier that bypasses lifestyle images to avoid unnecessary processing, achieving **under 2% misrouting** in regression tests.
- Architected a real-time LangGraph-based agentic **product-enrichment POC** using VLM's to generate swatch images and descriptive or inferred attributes, with a **seller-in-the-loop correction step**; deterministic evaluation achieved **~30%+ coverage improvement** over existing catalog (near **100% accuracy**)
- Monitored **3 production AI services** using developing automated scripts to track pipeline health and output quality

Data Scientist

Aug 2022 – Dec 2024

QuantumLink Communications, Mumbai

- Automated an **end-to-end customer engagement scoring** model for a B2B SaaS, with lead interest estimation model, sentiment signals from remarks, usage-based clustering and cohort analysis; sales teams used the scores to prioritize at-risk accounts, with **95%** of churned customers scoring below the engagement threshold in a **3-month retrospective**.
- Streamlined ETL pipelines (cron jobs on a Linux server) across **3 data sources** and built **5+** Power BI dashboards for cross-functional teams, eliminating 4+ hours/day of manual reporting.

Business Analyst

Jun 2021 – Feb 2022

Sciative Solutions, Navi Mumbai

- Redesigned pricing and auditing strategies for **3+** travel clients, projecting revenue **7% above** the clients' original forecasts; by building a pricing audit that scanned **6,000+** pricing cases per day against profitability issues for review.

TECHNICAL SKILLS

Machine Learning & Data Science: Python, SQL, TensorFlow/Keras, scikit-learn, Clustering, Feature Engineering, Model Evaluation, Statistical Analysis, Supervised learning, Unsupervised learning

Computer Vision & Image Processing: PyTorch, OpenCV, VLM's applications

Generative AI & LLM Applications: LangGraph, LangChain, Agentic Workflows, RAG, Vector Databases, LoRA Fine-tuning, MCP

Deployment & Tools: GCP (Vertex AI, BigQuery, Cloud Storage, Pub/Sub), FastAPI, Streamlit, Event-driven Pipelines, ETL Automation, Logging & Monitoring, Linux, Git, Docker (basic)

EDUCATION

M.Sc. Mathematics

2019 – 2021

Mithibai College, Mumbai

B.Sc. Mathematics

2016 – 2019

Jai Hind College, Mumbai